Smart Renewable Energy Management System with Adaptive Load Shedding

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***Abstract*—This paper presents a MATLAB/Simulink design of a solar photovoltaic microgrid for a residential block of 12 flats in Coimbatore, India. A 38.4-kW photovoltaic array with perturb-and-observe maximum power point tracking feeds a 400-V DC bus supported by a 130-kWh battery and a 10-F supercapacitor with low-pass-filter power sharing. A Stateflow load manager sheds and restores three priority tiers, and a five-level cascaded H-bridge inverter supplies the AC loads. Driven by NASA POWER weather data, the array supplies 113 percent of the 211.6-kWh daily demand on a clear day and 73 percent on a partly cloudy day. Over both full days the DC bus stays between 392.6 and 413.9 V, only the deferrable tier is shed, and the critical and essential loads are never interrupted. The inverter delivers 226 V RMS with a voltage total harmonic distortion of 0.13 percent up to the 50th harmonic, within IEEE Std 519.**

***Keywords*—Solar photovoltaic system, maximum power point tracking, DC microgrid, hybrid energy storage, supercapacitor, energy management system, adaptive load shedding, cascaded H-bridge inverter.**

I. Introduction

The transition towards low-carbon electricity has moved generation closer to the consumer, and residential microgrids built around rooftop solar photovoltaic (PV) arrays are now a practical way to supply apartment blocks. A microgrid groups local sources, storage and loads under a common controller so that the cluster can be operated as a single controllable unit [1], [2]. Its control is normally organised in layers, from fast converter-level loops to slower supervisory energy management [3], [4]. For a PV-dominated residential microgrid, the supervisory layer has to deal with one central difficulty: solar output follows irradiance and cell temperature, peaking near midday, whereas household demand peaks in the morning and, more strongly, in the evening.

At the source, the PV array must be operated at its maximum power point (MPP) while irradiance and temperature change. Accurate single-diode array models allow this behaviour to be reproduced in simulation [5], and maximum power point tracking (MPPT) algorithms adjust the operating voltage of a DC–DC converter to follow the MPP. Of the many MPPT methods available [6], perturb and observe (P&O) remains the most widely used because it needs only voltage and current measurements; its step size and sampling interval decide the trade-off between tracking speed and steady-state oscillation [7].

Storage is needed to move the midday surplus to the evening deficit. Battery energy storage systems (BESS) have high energy density but limited power density, and frequent high-rate current changes caused by cloud transients and load steps accelerate their ageing. Hybrid energy storage systems (HESS) that pair a battery with a supercapacitor address this by giving the slow part of the power mismatch to the battery and the fast part to the supercapacitor [8], [9]. Power sharing in a HESS has been obtained by optimisation [10], model predictive control [11], low-pass filtering with fixed or variable time constants [12], and by routing the high-frequency component of the DC-bus voltage controller, together with the uncompensated battery current, to the supercapacitor converter [13]. State-of-charge (SoC) and operating-mode logic is then added so that neither device is pushed beyond its limits [14], [15].

When generation and stored energy together cannot meet demand, the remaining option is to reduce the load in a controlled way. Residential loads differ widely in importance: refrigeration, communication and emergency lighting are critical, lighting and fans are essential, and air conditioners, water heaters and laundry are deferrable. A priority-based load manager can shed the lowest tier first and restore it when storage recovers, which keeps critical services supplied for longer than an undifferentiated cut-off. Load shedding of this kind is also an established part of islanded microgrid control [16]. Supervisory schemes of this kind have been combined with MPPT and storage control in hybrid and residential microgrids [17], [18].

Finally, the DC bus has to supply standard AC appliances. Multilevel inverters synthesise a stepped output from several DC sources and give lower harmonic distortion and lower device stress than two-level inverters [19], [20]. The cascaded H-bridge (CHB) topology is modular [21], and with phase-shifted carrier modulation [22] a five-level output can be obtained from two series-connected cells.

This work models these stages together for a block of 12

flats in Coimbatore, India. The main contributions are:

- an appliance-level, three-tier load model for 12 flats and a PV array sized against its firm daily energy, connected through a boost converter with P&O MPPT to a common $400\,\mathrm{V}$ DC bus and driven by NASA POWER hourly weather for a clear and a partly cloudy day;
- a battery–supercapacitor HESS with a low-pass-filter energy management system (EMS), SoC-aware power limits, and a combined current controller that passes fast DC-bus transients to the supercapacitor;
- a four-state Stateflow load manager with hysteresis, driven by the battery's own SoC, that sheds and restores load tiers in priority order;
- a five-level CHB inverter with a damped LC filter, evaluated up to the 50th harmonic; and
- a time-compressed integrated model of the complete microgrid, validated over two full 24-hour days.

II. Literature Summary

Hilmi *et al.* presented a finite-state-machine EMS for a standalone PV system with battery–supercapacitor storage, in which the operating mode is selected from the SoC of both devices [14]. Dwivedi and Ray studied energy management of a grid-connected microgrid with HESS, including power exchange and sharing between the battery and supercapacitor [15]. Khemissi *et al.* combined MPPT with a supervisory strategy in a wind/PV/battery smart microgrid [17], and Rajkumar and Gangatharan proposed an AC/DC residential microgrid with PV and storage that reduces unnecessary conversion stages [18]. On power sharing, Choi *et al.* formulated an optimisation that reduces battery-current fluctuation and supercapacitor loss [10], Wu *et al.* used low-pass filtering with a variable time constant [12], Hredzak *et al.* applied model predictive control [11], and Kollimalla *et al.* assigned the high-frequency part of the DC-link controller output and the uncompensated battery current to the supercapacitor [13]. Device models for the battery [23] and the double-layer capacitor [24] used in common simulation libraries are also well established.

Together, these studies cover PV integration, hybrid storage and supervisory control. The present project applies them to a specific residential design problem: a switching-level PV model driven by a weather profile, a tiered appliance-level demand model, SoC-aware HESS power allocation, and a priority load manager, all in one Simulink framework.

III. Research Gap

PV output is intermittent and does not coincide with residential demand. A battery-only interface is exposed to fast power variations, and a fixed load-shedding rule does not respond to changes in generation, demand or storage state. The literature reviewed above treats HESS power sharing, DC-bus regulation, MPPT and residential energy management mostly as separate problems. The gap addressed here is to bring these functions together in one simulation of a realistic residential block: filtering the power mismatch into slow and fast components, constraining battery operation by SoC, and coordinating any remaining deficit with a priority-based load manager.

The project objectives are to:

- model the residential demand of 12 flats by appliance and priority tier;
- model a PV source and its boost-converter interface, and track the MPP with P&O under a time-varying weather profile;
- connect PV generation to a regulated $400\,\mathrm{V}$ DC bus;
- integrate battery–supercapacitor storage with SoC-aware power sharing and priority load shedding; and
- evaluate generation, energy balance, DC-bus voltage, storage response, inverter output quality and load-tier management.

IV. Block Diagram Explanation

Fig. 1 shows the system architecture. On the generation path, irradiance and ambient-temperature profiles drive the PV array model, and the boost converter with P&O MPPT transfers the available PV power to a common $400\,\mathrm{V}$ DC bus. The design uses two identical $19.2\,\mathrm{kW}$ PV units ($38.4\,\mathrm{kW}$ in total). The PV unit test (Test 1) simulates one unit in detail, while the integrated model (Fig. 12) contains both units.

The battery and supercapacitor are connected to the DC bus in parallel through bidirectional DC–DC converters. The battery takes the slow, sustained part of the power mismatch and the supercapacitor the fast part. The DC bus feeds a five-level CHB inverter and an LC filter, which supply the AC loads of the 12 flats. A supervisory path uses PV power, load demand, DC-bus voltage and storage state: the EMS generates the storage references, and the Stateflow load manager sheds or restores load tiers when storage or power limits are reached. In the 24-hour integrated model, the inverter and AC loads are represented by an equivalent DC load (Section V-J); the switching inverter is evaluated separately in Test 2.

V. Methodology

The methodology consists of building the demand model, sizing and modelling the PV source, sizing and controlling the HESS, designing the supervisory EMS and load manager, modelling the CHB inverter, and evaluating each subsystem and the integrated model in MATLAB/Simulink. All parameters are set by the initialisation script `init_microgrid.m`.

A. Residential Load Model

The block contains $N = 12$ identical flats and a set of common loads. Each appliance a is described by its tier, quantity n_a, rated power P_a and a schedule of duty factors $\delta_a(h)$ for each hour h. The hourly demand of tier k is

$$P_k(h) = \sum_{a \in \mathcal{T}_k} n_a P_a \delta_a(h), \qquad k = 1, 2, 3, \tag{1}$$

and the daily tier energy is $E_k = \sum_{h=0}^{23} P_k(h) \times 1\,\mathrm{h}$. The schedule is calibrated so that each flat uses about $15.5\,\mathrm{kWh/day}$ on a summer design day, of which air conditioning is the largest share. Table I lists the tiers and Fig. 2 shows the

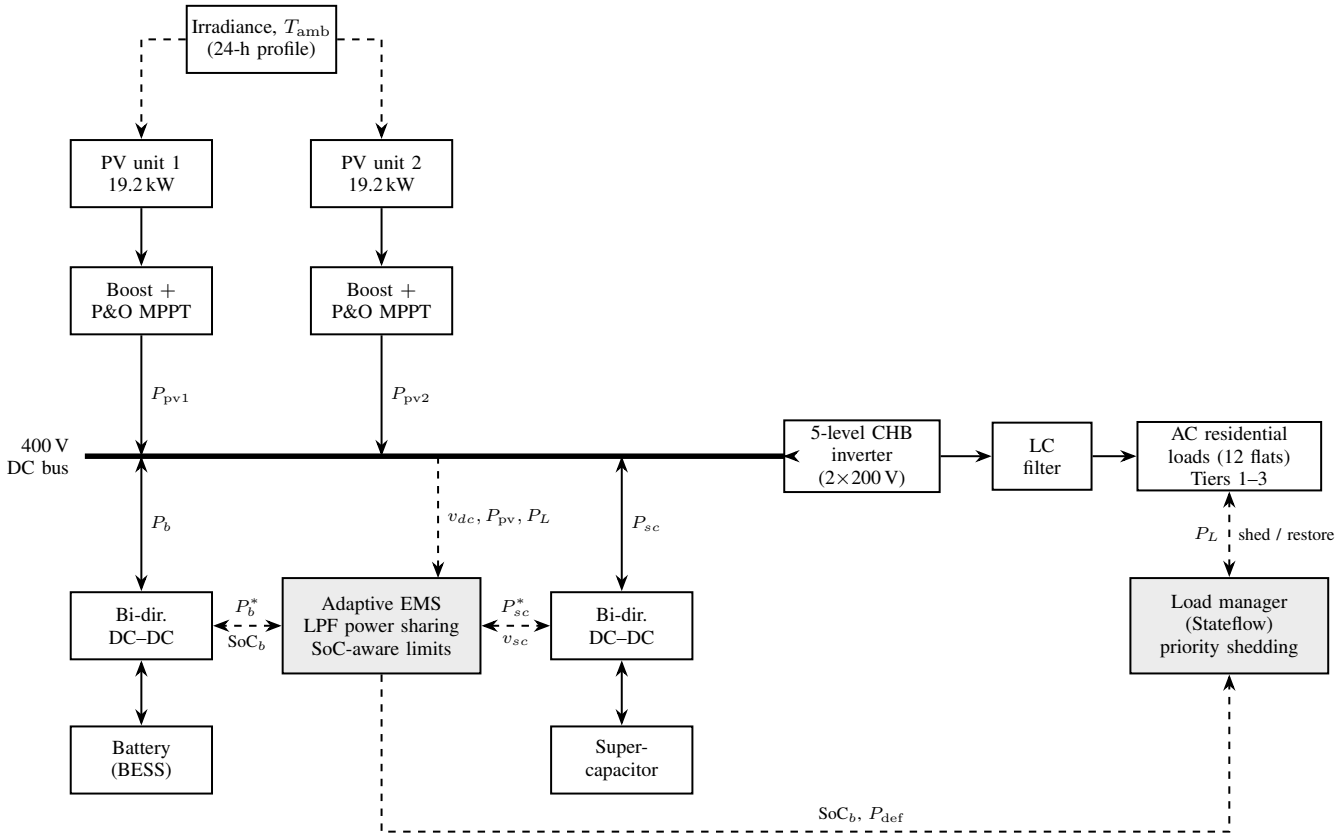


Fig. 1. System architecture. Solid lines denote power flow; dashed lines denote measurement and control signals. The integrated Simulink model contains both PV branches; the CHB inverter is simulated in detail separately (Test 2) and represented by an equivalent DC load in the 24-hour run.

TABLE I
RESIDENTIAL LOAD PRIORITY TIERS (12 FLATS)

Tier	Representative loads	Peak (kW)	Energy (kWh/d)
1 (critical)	Refrigerators, Wi-Fi routers, CCTV/security, common and emergency lighting	1.45	28.8
2 (essential)	Flat lighting, ceiling fans, TV and plug loads, kitchen appliances, lift	9.87	89.0
3 (deferrable)	Split air conditioners, geysers, washing machines, irons, water pump	9.00	93.8
Firm	Tiers 1–2	–	117.8
Total	Tiers 1–3	16.5[a]	211.6

[a]Coincident peak of the total profile (21:00–23:00).

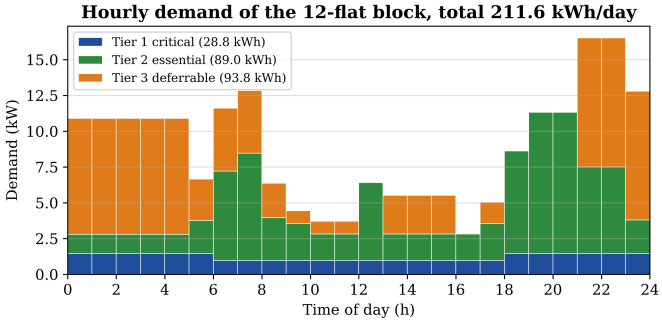


Fig. 2. Hourly demand of the 12-flat block by priority tier, computed from the appliance schedules in the initialisation script. Daily energy per tier is given in the legend.

resulting profile. The total demand is $211.6\,\mathrm{kWh/day}$, the firm (Tier 1 and Tier 2) demand is $117.8\,\mathrm{kWh/day}$, and the peak is $16.5\,\mathrm{kW}$ between 21:00 and 23:00, when PV output is zero. The firm demand between 18:00 and 06:00 is $67.8\,\mathrm{kWh}$, which sets the overnight storage requirement.

B. Photovoltaic Array Sizing

The PV array is sized to supply the firm daily energy. Using a peak-sun-hour value of $\mathrm{PSH} = 4.8\,\mathrm{h/day}$, a temperature derating factor $\eta_{\mathrm{temp}} = 0.88$, a soiling, mismatch and wiring factor $\eta_{\mathrm{soil}} = 0.95$ and a system factor $\eta_{\mathrm{sys}} = 0.85$, the minimum rating is

$$P_{\mathrm{PV,min}} = \frac{E_{\mathrm{firm}}}{\mathrm{PSH}\,\eta_{\mathrm{temp}}\,\eta_{\mathrm{soil}}\,\eta_{\mathrm{sys}}} = \frac{118}{4.8 \times 0.88 \times 0.95 \times 0.85} \approx 34.6\,\mathrm{kW}. \tag{2}$$

An installed rating of $38.4\,\mathrm{kW}$ is selected, which gives an 11% margin over this estimate.

TABLE II
PV ARRAY AND BOOST-CONVERTER PARAMETERS

Parameter	Value
Module rating / V_{mp} / I_{mp}	400 W / 40 V / 10 A
Module V_{oc} / I_{sc}	48 V / 10.5 A
Series modules × parallel strings	6 × 8 per unit
PV rating per unit / total (two units)	19.2 kW / 38.4 kW
Nominal string V_{mp} / V_{oc}	240 V / 288 V
DC-bus reference	400 V
Boost inductance	0.30 mH
Input / output capacitance	270 µF / 270 µF
Switching frequency	20 kHz
MPPT method / duty step ΔD	P&O / 0.002
Duty-cycle window (MPPT mode)	0.36–0.60
Night parking	$D = 0.42$ when $P_{\mathrm{pv}} < 50$ W

C. Module and Array Configuration

Each module is rated 400 W with $V_{\mathrm{mp}} = 40$ V, $I_{\mathrm{mp}} = 10$ A, $V_{\mathrm{oc}} = 48$ V, $I_{\mathrm{sc}} = 10.5$ A and 72 cells. Six modules are connected in series per string and eight strings in parallel per unit, so

$$P_{\mathrm{unit}} = 6 \times 8 \times 400\,\mathrm{W} = 19.2\,\mathrm{kW}, \tag{3}$$

$$V_{\mathrm{mp,string}} = 6 \times 40\,\mathrm{V} = 240\,\mathrm{V}, \tag{4}$$

$$V_{\mathrm{oc,string}} = 6 \times 48\,\mathrm{V} = 288\,\mathrm{V}, \tag{5}$$

with a nominal unit current of $8 \times 10\,\mathrm{A} = 80\,\mathrm{A}$. The array is represented by the single-diode PV Array block, whose fitted parameters include a series resistance of $0.200\,\Omega$ and a shunt resistance of $821.8\,\Omega$ per module, and temperature coefficients of $-0.361\%/^{\circ}\mathrm{C}$ for V_{oc} and $+0.102\%/^{\circ}\mathrm{C}$ for I_{sc} [5]. These ratings refer to standard test conditions (STC); under the weather profile the output is lower because irradiance is below $1000\,\mathrm{W/m^2}$ and the cells are hotter than $25\,^{\circ}\mathrm{C}$. Table II summarises the PV-side parameters.

D. Weather and Cell-Temperature Model

The PV model is driven by weather data from the NASA Prediction Of Worldwide Energy Resources (POWER) database [29] for Coimbatore (11.02° N, 76.96° E). Hourly all-sky global horizontal irradiance (`ALLSKY_SFC_SW_DWN`) and 2 m air temperature (`T2M`) were downloaded for March–May 2025, and two design days were chosen from the daily totals: 9 May 2025, a clear day with $7.32\,\mathrm{kWh/m^2}$, and 16 May 2025, a partly cloudy day with $4.65\,\mathrm{kWh/m^2}$. The hourly values are placed at the hour midpoints and interpolated with a shape-preserving piecewise cubic (PCHIP) to one-minute resolution, and the resulting profiles are applied through look-up tables. On the clear day the irradiance peaks at $1002\,\mathrm{W/m^2}$ at 11:30 and the ambient temperature ranges from 22.1 to $35.9\,^{\circ}\mathrm{C}$; on the partly cloudy day the irradiance peaks at $790\,\mathrm{W/m^2}$ and falls sharply after 14:00. The cell temperature is estimated from the ambient temperature and irradiance G with the nominal operating cell temperature (NOCT) relation

$$T_{\mathrm{cell}} = T_{\mathrm{amb}} + \frac{\mathrm{NOCT} - 20}{800}\,G, \qquad \mathrm{NOCT} = 45\,^{\circ}\mathrm{C}, \tag{6}$$

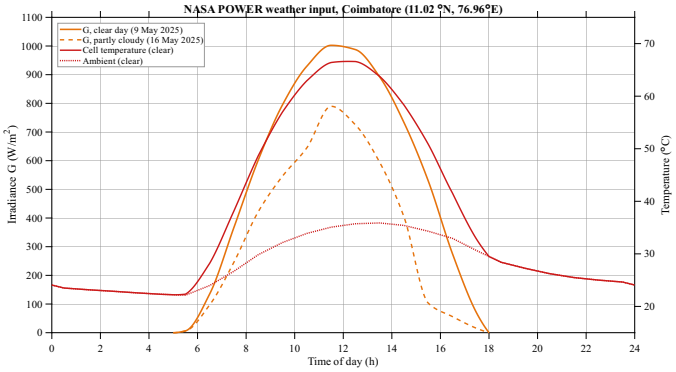


Fig. 3. NASA POWER weather input for Coimbatore: irradiance on the clear (9 May 2025) and partly cloudy (16 May 2025) days, with the ambient and modelled cell temperature for the clear day.

with temperatures in °C and G in $\mathrm{W/m^2}$. Fig. 3 shows the inputs. For a quick check of the simulated output, the MPP power is approximated as

$$P_{\mathrm{MPP}} \approx P_{\mathrm{STC}}\,\frac{G}{1000}\,[1 + \gamma\,(T_{\mathrm{cell}} - 25)]\,, \tag{7}$$

where $\gamma = -0.004/^{\circ}\mathrm{C}$ is the power temperature coefficient.

E. Boost Converter and Maximum Power Point Tracking

The string MPP voltage is about 240 V at STC while the DC bus is held at 400 V, so a boost converter is used [25]. In continuous conduction the PV voltage and duty ratio are related by

$$V_{\mathrm{pv}} = (1 - D)\,V_{\mathrm{dc}}, \qquad D_{\mathrm{STC}} = 1 - \frac{240}{400} = 0.40. \tag{8}$$

Lowering D therefore raises the PV voltage. The P&O algorithm [7] samples V_{pv} and I_{pv} every 0.1 ms, computes $\Delta P_k = P_k - P_{k-1}$ and $\Delta V_k = V_k - V_{k-1}$, and updates the duty ratio as

$$D_{k+1} = \begin{cases} D_{\mathrm{park}} = 0.42, & P_k < 50\,\mathrm{W}, \\ D_k, & \Delta P_k = 0, \\ D_k - \Delta D, & \Delta P_k\,\Delta V_k > 0, \\ D_k + \Delta D, & \text{otherwise}, \end{cases} \tag{9}$$

with $\Delta D = 0.002$, and then limits D_{k+1} to 0.36–0.60. This window corresponds to PV voltage commands of 160–256 V. It covers every MPP of the day (about 195–245 V) but stays below the open-circuit voltage, so the tracker cannot drift to open circuit when irradiance rises at sunrise. The night parking state holds the operating point near $0.8\,V_{\mathrm{oc}}$ when the PV power is negligible.

The PV block also includes a curtailment mode for a full battery. A relay with hysteresis switches the PV converter from MPPT to a DC-bus voltage PI controller ($K_p = 0.001$, $K_i = 0.05$, duty limited to 0.05–0.85) when the battery SoC rises above 98%, and returns it to MPPT when the SoC falls below 95%. The hysteresis band prevents the converter from chattering between the two modes. A 20 kHz sawtooth carrier compared with the duty command generates the gate signal.

F. Hybrid Energy Storage Sizing and Modelling

1) Battery: The battery supplies the firm overnight demand plus a reserve. With $E_{\text{night}} \approx 68\,\text{kWh}$, a reserve $E_{\text{res}} = 30\,\text{kWh}$, a usable depth of discharge of 0.8 and a discharge efficiency $\eta_d = 0.95$,

$$E_{\text{bat}} = \frac{E_{\text{night}} + E_{\text{res}}}{\text{DoD}\,\eta_d} = \frac{68 + 30}{0.8 \times 0.95} \approx 129\,\text{kWh}, \qquad (10)$$

and a 130 kWh, 240 V lithium-ion bank (541.7 Ah) is chosen. Within the 20–90% SoC operating window this leaves 91 kWh of usable energy, which exceeds the firm night demand. The battery is represented by the generic lithium-ion block based on the model of Tremblay and Dessaint [23], with an internal resistance of $0.05\,\Omega$ and an initial SoC of 80%.

Because a whole day is compressed into a few simulated seconds (Section V-L), the SoC must move as far in one compressed day as it would in one real day. A time-scaled capacity cannot simply be entered in the battery block: the block derives its internal resistance and polarisation constants from the rated capacity, so a capacity of a few milliampere-hours would give a resistance of hundreds of ohms and an unrealistic terminal voltage. The block therefore keeps the real capacity of 541.7 Ah, so its voltage and resistance are physical, and the SoC used by the EMS and the load manager is computed by a separate coulomb counter,

$$\text{SoC}_b(t) = \text{SoC}_b(0) - \frac{100}{3600\,Q_{\text{sim}}} \int_0^t i_b(\tau)\,d\tau, \qquad (11)$$

with the time-scaled capacity

$$Q_{\text{sim}} = Q_{\text{real}}\,\frac{t_{\text{hour}}}{3600} = 541.7 \times \frac{0.12}{3600} = 0.0181\,\text{Ah}, \qquad (12)$$

so that the SoC swing over one compressed day equals that over one real day.

2) Supercapacitor: The supercapacitor is designed as a 10 F, 250 V bank. Its stored energy is

$$E_{sc} = \tfrac{1}{2} C_{sc} v_{sc}^2 = \tfrac{1}{2} \times 10 \times 250^2 = 312.5\,\text{kJ}. \qquad (13)$$

Discharging to half of the rated voltage releases 75% of this energy, about 234 kJ, enough to supply a 20 kW transient for roughly 12 s. It is represented by the Stern-type supercapacitor block [24].

3) DC link and converters: The DC-link capacitance is sized to supply a 20 kW step for 10 ms with a 5% voltage dip:

$$C_{dc} = \frac{2P\,\Delta t}{V_{\text{dc}}^2 - (0.95 V_{\text{dc}})^2} = \frac{2 \times 20\,000 \times 0.01}{400^2 - 380^2} \approx 25.6\,\text{mF}. \qquad (14)$$

Each storage element connects to the bus through a non-isolated bidirectional half-bridge converter with two IGBT/diode switches: 0.30 mH for the battery and 0.355 mH for the supercapacitor. Table III lists the storage and DC-link parameters. Every block takes its value directly from the initialisation script.

TABLE III
STORAGE AND DC-LINK PARAMETERS

Parameter	Value
Battery type / nominal voltage	Li-ion / 240 V
Capacity in battery block	541.7 Ah (130 kWh)
SoC used by EMS / load manager	Coulomb counter, $Q_{\text{sim}} = 0.0181$ Ah
Battery current limit	± 156 A
Initial SoC / window	80% / 20–90%
SC capacitance	10 F
SC rated / initial voltage	250 V / 250 V
SC current limit	± 200 A
DC-link capacitance	25 mF
Storage converter PWM	20 kHz
LPF time constant τ	2 ms (1 real minute)

G. Adaptive Energy Management System

The EMS works on the power mismatch between demand and PV generation,

$$P_{\text{mis}}(t) = P_L(t) - P_{\text{pv}}(t), \qquad (15)$$

which is positive in deficit and negative in surplus. A first-order discrete low-pass filter extracts its slow component for the battery:

$$P_{b,\text{base}}[k] = \alpha\,P_{b,\text{base}}[k-1] + (1-\alpha)\,P_{\text{mis}}[k], \quad \alpha = e^{-T_s/\tau}, \qquad (16)$$

where T_s is the sample time and τ the filter time constant. An adaptive function then derates the battery request near its SoC limits:

$$P_b^* = \begin{cases} 0.2\,P_{b,\text{base}}, & \text{SoC}_b < 30\% \text{ and } P_{b,\text{base}} > 0, \\ 0.2\,P_{b,\text{base}}, & \text{SoC}_b > 90\% \text{ and } P_{b,\text{base}} < 0, \\ P_{b,\text{base}}, & \text{otherwise.} \end{cases} \qquad (17)$$

The supercapacitor receives the remainder, $P_{sc}^* = P_{\text{mis}} - P_b^*$, which contains the fast part of the mismatch and any battery power withheld by (17). The power references are converted to current references,

$$i_b^* = \frac{P_b^*}{v_b}, \qquad i_{sc,\text{EMS}}^* = \frac{P_{sc}^*}{v_{sc}} - i_{sc}, \qquad (18)$$

with the measured voltages limited to 1–1000 V to avoid division by zero. The unfiltered mismatch is also passed to the load manager as the power deficit P_{def}. Table IV summarises the resulting operating modes.

H. Combined Battery/Supercapacitor Current Control

The EMS references are tracked by a combined controller that also regulates the DC bus, following the structure of [13]. A PI controller on the bus-voltage error $e_v = V_{\text{dc}}^* - v_{\text{dc}}$ gives the total current demand i_{tot}^*. A first-order low-pass filter extracts its slow part $\bar{i}_{\text{tot}}^*$, and the high-frequency remainder is added to the uncompensated battery current error and referred to the supercapacitor side:

$$i_{sc}^* = \left[(i_{\text{tot}}^* - \bar{i}_{\text{tot}}^*) + (i_b^* - i_b)\right] \frac{v_b}{v_{sc}} + i_{sc,\text{EMS}}^*. \qquad (19)$$

TABLE IV
EMS OPERATING MODES

Mode	Condition	Action
M1	PV surplus, $\mathrm{SoC}_b < 98\%$	Charge storage, charging derated above 90% SoC
M2	PV surplus, $\mathrm{SoC}_b \geq 98\%$	PV switches to DC-bus voltage control (curtailment) until $\mathrm{SoC}_b < 95\%$
M3	Deficit, storage available	Battery supplies slow component; SC supplies residual
M4	Deficit, $\mathrm{SoC}_b < 30\%$	Battery discharge derated to 20%; load manager sheds tiers
M5	PV negligible (night)	MPPT parked; storage supplies loads within limits

TABLE V
CONTROLLER GAINS AND LIMITS

Loop	K_p	K_i
DC-bus voltage (outer)	5.03	63.2
Battery current (inner)	0.0094	11.8
Supercapacitor current (inner)	0.0112	14.0
PV constant-voltage (curtailment)	0.001	0.05
Current limits: battery ± 156 A, supercapacitor ± 200 A		
Duty ratio 0–0.9, clamping anti-windup		

The battery current loop tracks the slow reference only, so fast bus disturbances are carried by the supercapacitor, and any current the battery cannot follow is passed to the supercapacitor as well. The battery reference is limited to $\pm 1.5 P_{b,\max}/V_{b,\mathrm{nom}} = \pm 156\,\mathrm{A}$ and the supercapacitor reference to ± 200 A. Each current PI output is limited to a duty ratio of 0–0.9 with clamping anti-windup and drives a 20 kHz PWM generator that switches the two converter legs complementarily.

Limiting the duty ratio below 1 prevents the converter from short-circuiting the battery during large transients. The gains (Table V) are the analytical values from the initialisation script: $K_{p,i} = 2\pi f_{ci} L/V_{\mathrm{dc}}$ and $K_{i,i} = K_{p,i}\, 2\pi f_{ci}/10$ with $f_{ci} = f_{sw}/10 = 2\,\mathrm{kHz}$ for the current loops, and $K_{p,v} = 2\pi f_v C_{dc} V_{\mathrm{dc}}/V_{sc,\mathrm{nom}}$, $K_{i,v} = K_{p,v}\, 2\pi f_v/10$ with $f_v = 20\,\mathrm{Hz}$ for the voltage loop.

I. Cascaded H-Bridge Inverter

A single-phase five-level CHB inverter [19], [20] is formed by two H-bridge cells (eight IGBT/diode switches in total), each supplied from a separate 200 V DC source. With the cell outputs in series, the output takes the levels -400, -200, 0, $+200$ and $+400$ V:

$$v_o = \sum_{j=1}^{2} S_j\, V_{\mathrm{dc},j}, \qquad S_j \in \{-1, 0, +1\}. \tag{20}$$

Gate signals are produced by unipolar phase-shifted carrier PWM [22]: a 50 Hz sinusoidal reference with modulation index $m = 0.8$ is compared with 1 kHz triangular carriers, the carrier of the second cell being shifted by 90°, so the output ripple appears at an effective 4 kHz. The output filter has $L_f = 2\,\mathrm{mH}$ ($0.05\,\Omega$) and $C_f = 20\,\mu\mathrm{F}$ with a damping resistor $R_d = 2\,\Omega$ in series with the capacitor [27]:

$$f_{\mathrm{res}} = \frac{1}{2\pi\sqrt{L_f C_f}} = 796\,\mathrm{Hz}, \tag{21}$$

which lies well above the fundamental and a factor of five below the effective switching frequency, so the switching harmonics are attenuated and do not excite the resonance. The test load is $11.8\,\Omega$, which draws 4.3 kW at 226 V RMS.

J. Load Representation in the Integrated Model

Simulating 50 Hz switching over a compressed day would add computation without changing the energy balance, so in the integrated model the inverter and the AC loads are represented by a controlled current source on the DC bus,

$$i_L = \frac{u_1 P_1(h) + u_2 P_2(h) + u_3 P_3(h)}{\eta_{\mathrm{inv}}\, v_{\mathrm{dc}}}, \qquad \eta_{\mathrm{inv}} = 0.96, \tag{22}$$

where $P_k(h)$ is the tier demand from (1) and u_k the connection command of the load manager. The loads are therefore resistive (unity power factor), and the daily demand seen by the DC bus is $211.6/0.96 = 220.4\,\mathrm{kWh}$.

K. Priority-Based Adaptive Load Shedding

The load manager is a Stateflow chart with four states. Its output L selects which tiers are connected (Table VI): tier k is connected when $L \leq 3-k$, that is $u_1 = (L \leq 2)$, $u_2 = (L \leq 1)$ and $u_3 = (L = 0)$. The transitions use the battery SoC (as a fraction), its rate of change, and the power deficit P_{def} from the EMS. The thresholds are $\theta_1 = 0.40$, $\theta_2 = 0.30$, $\mathrm{SoC}_{\min} = 0.20$ and a restoration hysteresis $d = 0.08$. Table VII gives the guards, and Fig. 4 shows the chart. The chart is driven by the coulomb-counted battery SoC of (11), converted from percent to a fraction, and P_{def} is converted from watts to kilowatts before it reaches the guards. The output L reaches the load block through a global signal tag and a one-step (2 ms) delay, which breaks the algebraic loop formed by load, bus, SoC and chart.

Shedding is anticipatory: the deferrable tier is dropped not only when SoC falls below θ_1, but also when it is within 0.05 of θ_1 and falling faster than 0.15 per unit time. Every restoration requires the SoC to recover by d above the shedding threshold, which prevents chattering around a threshold. The load function also tracks the unserved deferrable energy,

$$E_{\mathrm{def}} \leftarrow \min\Big(E_{\max},\; E_{\mathrm{def}} + P_3(1-u_3)\,\Delta h - u_3\,\frac{E_{\mathrm{def}}}{T_c}\,\Delta h\Big), \tag{23}$$

where Δh is the real time per solver step, $T_c = 2\,\mathrm{h}$ is the recovery time constant and $E_{\max} = 10\,\mathrm{kWh}$. This records how much deferred demand must be served after the tier is restored. Random variability of 3%, 10% and 15% can be added to Tiers 1, 2 and 3 respectively.

TABLE VI
LOAD-MANAGER STATES

State	Tier 1	Tier 2	Tier 3
$L = 0$ Normal	On	On	On
$L = 1$ Shed deferrable	On	On	Off
$L = 2$ Critical only	On	Off	Off
$L = 3$ Outage	Off	Off	Off

TABLE VII
LOAD-MANAGER TRANSITION GUARDS

Transition	Guard
$0 \to 1$	$\mathrm{SoC} < \theta_1$ or ($\mathrm{SoC} < \theta_1 + 0.05$ and $\dot{\mathrm{SoC}} < -0.15$)
$1 \to 2$	$\mathrm{SoC} < \theta_2$
$2 \to 3$	$\mathrm{SoC} \le \mathrm{SoC}_{\min}$ or $P_{\mathrm{def}} > 22$
$3 \to 2$	$\mathrm{SoC} > \mathrm{SoC}_{\min} + 0.10$ and $P_{\mathrm{def}} \le 10$
$2 \to 1$	$\mathrm{SoC} > \theta_2 + d$
$1 \to 0$	$\mathrm{SoC} > \theta_1 + d$

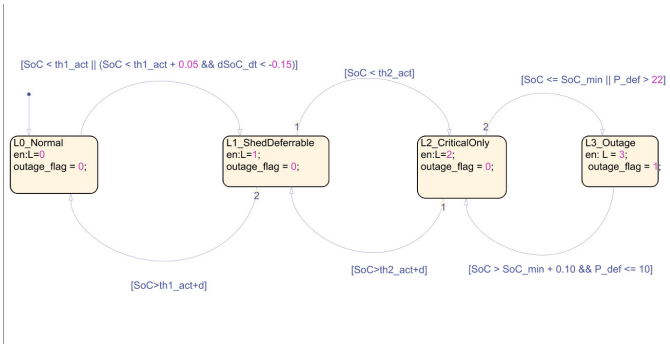


Fig. 4. Stateflow load-management chart with normal, deferrable-shedding, critical-only and outage states.

L. Simulation Procedure and Performance Evaluation

All power stages are simulated in discrete mode with a fixed step of $1\,\mu\mathrm{s}$. A full switching simulation of 24 real hours is not practical, so time is compressed: t_{hour} simulated seconds represent one real hour. The PV unit model uses $t_{\mathrm{hour}} = 0.05\,\mathrm{s}$ ($1.2\,\mathrm{s}$ per day). The integrated model uses $t_{\mathrm{hour}} = 0.12\,\mathrm{s}$ ($2.88\,\mathrm{s}$ per day), runs the EMS and Stateflow chart every $2\,\mathrm{ms}$ (one real minute), and updates loads and weather every $30\,\mathrm{ms}$ (15 real minutes). Table VIII lists the test conditions.

The following indices are used. The daily PV energy is $E_{\mathrm{pv}} = \int P_{\mathrm{pv}}\,dt$ expressed in real hours, and the energy coverage ratio is $\rho = E_{\mathrm{pv}}/E_{\mathrm{load}}$. The DC-bus deviation is

$$\Delta V_{\mathrm{dc}}(\%) = \frac{\max |v_{\mathrm{dc}} - V_{\mathrm{dc}}^{*}|}{V_{\mathrm{dc}}^{*}} \times 100. \tag{24}$$

Inverter output quality is measured by the total harmonic distortion,

$$\mathrm{THD} = \frac{\sqrt{\sum_{h=2}^{\infty} X_h^2}}{X_1} \times 100\%, \tag{25}$$

where X_h is the RMS value of the h-th harmonic of voltage or current, and the result is compared with IEEE Std 519 [26].

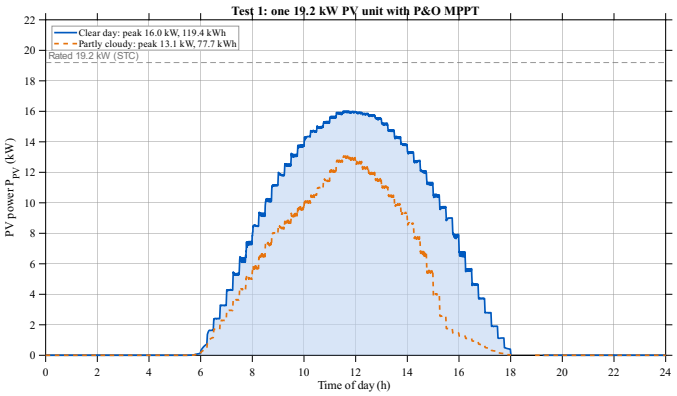


Fig. 5. Test 1: power of one $19.2\,\mathrm{kW}$ PV unit with P&O MPPT on the clear and partly cloudy days.

VI. RESULTS AND DISCUSSION

This section reports the tests in Table VIII. Table X at the end of the section collects the quantitative indices.

A. Test 1: Photovoltaic Generation under Measured Weather

Fig. 3 shows the applied weather. On the clear day the irradiance rises after 05:30, peaks at $1002\,\mathrm{W/m^2}$ at 11:30 and returns to zero at 18:00. The cell temperature from (6) follows the irradiance and reaches $66.6\,^{\circ}\mathrm{C}$ shortly after noon, while at night it falls back to the ambient level of 22–25 °C. On the partly cloudy day the irradiance peaks at $790\,\mathrm{W/m^2}$ and collapses after 14:00.

Fig. 5 compares the power of one PV unit on the two days. On the clear day the unit peaks at $16.0\,\mathrm{kW}$ at 11:38, which is 83% of the $19.2\,\mathrm{kW}$ STC rating. This agrees with (7): at $G \approx 1000\,\mathrm{W/m^2}$ and $T_{\mathrm{cell}} \approx 66\,^{\circ}\mathrm{C}$,

$$P_{\mathrm{MPP}} \approx 19.2 \times 1.00 \times [1 - 0.004\,(66 - 25)] \approx 16.0\,\mathrm{kW}. \tag{26}$$

The daily energy is $119.4\,\mathrm{kWh}$ per unit. On the partly cloudy day the unit peaks at $13.1\,\mathrm{kW}$ at 11:35 and produces $77.7\,\mathrm{kWh}$, 35% less.

Fig. 6 shows the PV voltage, current and duty ratio on the clear day, and each part of the P&O logic in (9) can be seen. At night the duty ratio is parked at 0.42. At dawn and dusk it reaches the lower limit of 0.36, where the tracker raises the PV voltage to find the MPP at low irradiance. During the day it rises from about 0.43 in the morning to 0.51 at noon. The PV voltage falls to $197\,\mathrm{V}$ at noon as the cells heat, and (8) gives $1 - 197/400 = 0.51$, matching the simulated duty ratio. The PV current reaches $83\,\mathrm{A}$, close to the nominal $80\,\mathrm{A}$. The DC-bus voltage in this single-unit test (Fig. 7) rises with PV power to at most 1.5% above $400\,\mathrm{V}$, which reflects the finite stiffness of the bus in the stand-alone test rather than a tracking error. The thicker bands in the traces are the P&O perturbation of $\pm\Delta D$ around the MPP.

B. Daily Energy Balance

Fig. 8 compares the two-unit PV profile with the demand. On the clear day the array peaks at about $32\,\mathrm{kW}$ and delivers

TABLE VIII
TEST CONDITIONS USED FOR THE SIMULATION RESULTS

Parameter	**Test 1 – PV unit**	**Test 2 – CHB inverter**	**Tests 3 and 4 – Load manager and integrated model**
Model	One 19.2 kW unit, boost, P&O	Two-cell CHB with damped LC filter	Full microgrid: two PV units, HESS, EMS, Stateflow load manager, DC-equivalent load
Time step	1 µs	1 µs	1 µs power, 2 ms EMS and chart, 30 ms loads
Time mapping	0.05 s per real hour	Real time	0.12 s per real hour
Duration	1.2 s (24 h)	0.1 s (5 cycles, last 2 analysed)	2.88 s (24 h), each day
Input	NASA POWER clear and partly cloudy days	200 V per cell; 11.8 Ω and 200 Ω loads	NASA POWER clear and partly cloudy days; 12-flat tier profile; initial SoC 80%
Signals recorded	V_{pv}, I_{pv}, P_{pv}, D, v_{dc}	v_o, i_o, inverter voltage; FFT to the 50th harmonic and to 10 kHz	v_{dc}, v_b, i_b, SoC$_b$, v_{sc}, P_{pv}, P_L, L

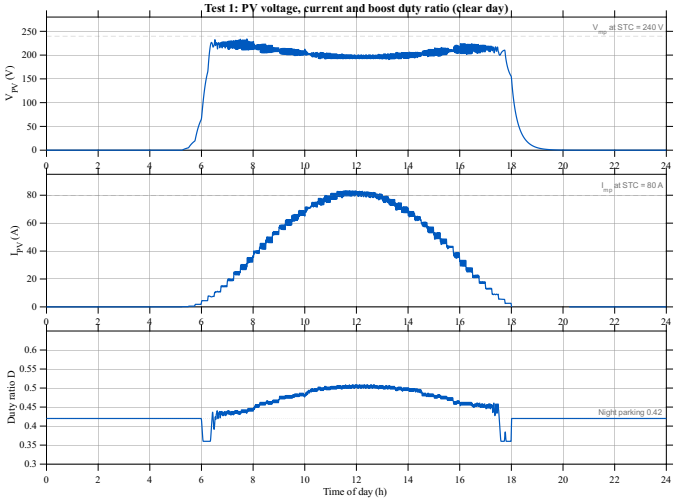


Fig. 6. Test 1: PV voltage, PV current and boost-converter duty ratio of one unit over the clear day.

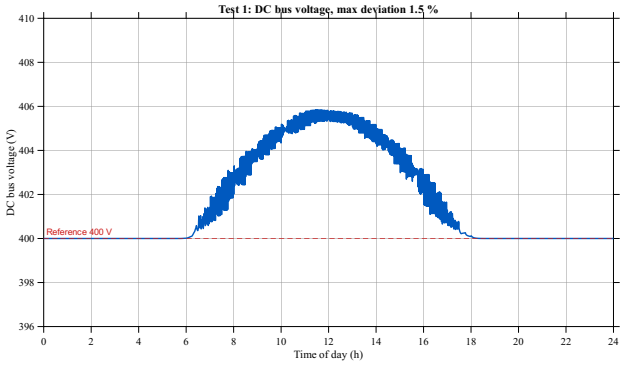


Fig. 7. Test 1: DC-bus voltage of the stand-alone PV unit test on the clear day.

238.9 kWh, against 211.6 kWh of demand, so the coverage ratio is $\rho = 1.13$. On the partly cloudy day it delivers 155.3 kWh ($\rho = 0.73$). Even on the partly cloudy day the array exceeds the 117.8 kWh firm demand by 32%.

The timing of the two curves matters more than the totals. On the clear day PV generation exceeds the total demand only between about 07:40 and 16:40, and the midday surplus reaches about 28 kW. Demand reaches its 16.5 kW peak at

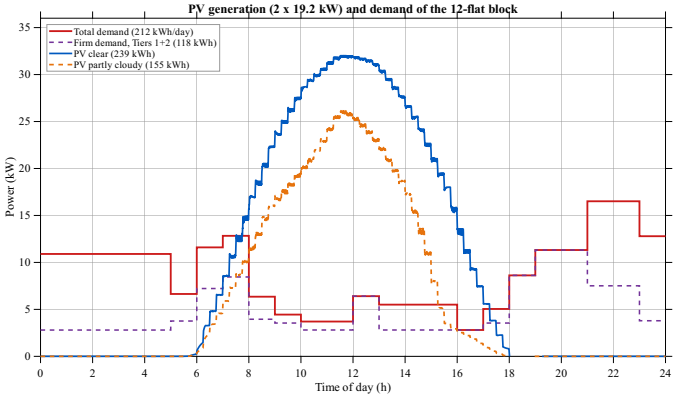


Fig. 8. Two-unit PV generation on the clear and partly cloudy days compared with the total and firm (Tier 1 + Tier 2) demand of the 12-flat block.

21:00–23:00, after sunset, and the whole night load must come from storage. The midday surplus must therefore be stored to supply the 67.8 kWh of firm night demand, which the 91 kWh usable battery capacity from (10) can cover. Around noon the surplus also exceeds the 25 kW battery power rating, and once the battery is full the PV output has to be curtailed. Deferrable night loads, mainly air conditioning between 21:00 and 05:00, are not covered by the battery sizing; they are left to the load manager, which is tested in Section VI-D.

C. Test 2: Cascaded H-Bridge Inverter

Fig. 9 shows the inverter output over two cycles at the 4.3 kW load. The unfiltered voltage shows the five levels 0, ±200 and ±400 V of (20), with PWM switching between adjacent levels at the effective 4 kHz. The filtered voltage is sinusoidal with a fundamental of 319.3 V peak (225.8 V RMS) and a highest instantaneous value of 325.7 V. With the resonance placed at 796 Hz, well below the switching frequency, the filter shows no resonant overshoot. The load current (Fig. 10) is sinusoidal with 19.1 A RMS and a small ripple at the switching frequency.

Fig. 11 shows the voltage harmonics up to the 50th order, the range counted by IEEE Std 519 [26]. The largest component is the 9th harmonic at less than 0.09% of the fundamental, and the voltage THD from (25) is 0.13%, against limits of 5% for an individual harmonic and 8% for THD at or below 1 kV. The current THD is also 0.13%, because the load is resistive.

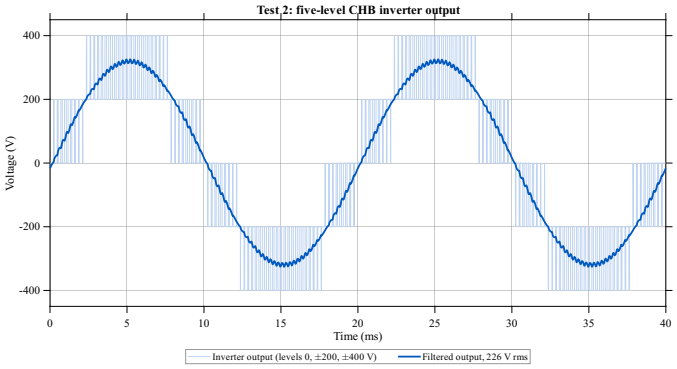


Fig. 9. Test 2: five-level inverter voltage (light) and filtered output voltage (dark) over two 50 Hz cycles at the 4.3 kW load.

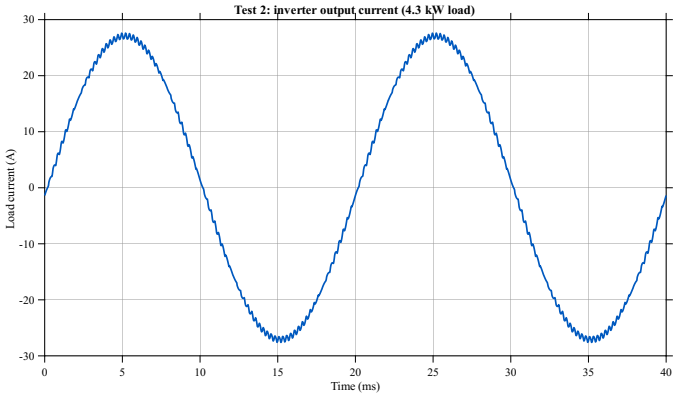


Fig. 10. Test 2: inverter output current at the 4.3 kW load.

TABLE IX
TEST 2: INVERTER RESULTS FOR THREE CASES

	Damped, full load	**Undamped, full load**	**Damped, light load**
R_d / R_{load} (Ω)	2 / 11.8	0 / 11.8	2 / 200
Output (V RMS) / power (kW)	225.8 / 4.32	225.8 / 4.32	227.1 / 0.26
Peak output voltage (V)	325.7	324.2	329.7
THD_v to 50th (%)	0.13	0.14	0.23
THD_v to 10 kHz (%)	1.57	1.30	1.85
THD_i to 50th (%)	0.13	0.14	0.23

Table IX compares three cases. Evaluated up to 10 kHz, which includes the switching sidebands, the voltage THD is 1.57% after the filter, compared with 34% for the unfiltered five-level waveform. Without the damping resistor the results at full load are nearly the same, because the resonance is not excited; the resistor is kept as a margin against resonance at light load, where the voltage peak rises only to 329.7 V. The resistor slightly reduces the attenuation at high frequency, which is why the wideband THD is 1.57% with damping and 1.30% without.

D. Tests 3 and 4: Integrated 24-Hour Operation

Fig. 12 shows the top level of the integrated model. The power stage (PV array, DC link, battery and supercapacitor converters, and the DC-equivalent load) is in the upper row, and the EMS, converter control, load manager, load profile and monitoring blocks, which exchange signals through global tags, are in the lower row. The load manager is driven by the battery's own SoC, so Tests 3 (load manager) and 4 (integrated model) are evaluated together in one closed-loop run over a full day. The model was run for both NASA POWER days, starting at midnight with an SoC of 80%.

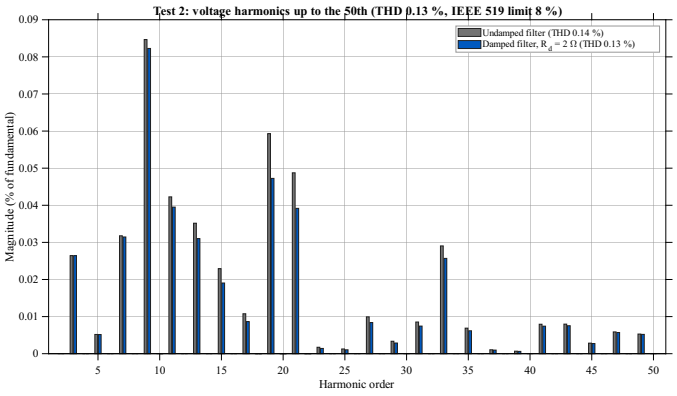


Fig. 11. Test 2: output-voltage harmonics up to the 50th order with the undamped and damped filter, in percent of the fundamental.

Clear day. Figs. 13–16 show the clear-day results. In the first simulated half hour the DC link and supercapacitor charge from their initial values while the battery works at its 156 A limit; because of the time compression this short transient is booked as real hours and costs about 7% SoC. After this the battery supplies the night load at about 45 A. At 03:40 the SoC falls below $\theta_1 = 40\%$, the load manager moves to $L = 1$ and Tier 3 is disconnected: the DC-side load drops from 11.4 kW to 2.9 kW and the battery current to about 12 A. The SoC reaches its minimum of about 32% near 07:10, above θ_2, so Tier 2 is never shed. PV power starts at 06:00, the battery is charging from about 07:30 at up to about 100 A near noon, and at 09:20 the SoC exceeds $\theta_1 + d = 48\%$ and Tier 3 is reconnected.

At 12:30 the battery reaches 98% and the PV converter switches to curtailment. Because the storage converters already hold the bus at 400 V, the PV output then falls almost to zero and the battery supplies the load until its SoC drops to 95%, when MPPT resumes. This happens three times between 12:30 and 16:00, with clean transitions and no chattering. In the evening the battery supplies the 16.5 kW peak (17.2 kW on the DC side) at about 65 A, and the day ends at an SoC of 39.1%; Tier 3 is shed again at about 23:50.

After start-up, the DC bus stays between 392.6 V and 413.9 V (Fig. 15). For most of the day it is within 1% of 400 V. The largest excursions occur at step changes in the battery charging power. At about 12:00 the SoC reaches 90%, the EMS derates the battery charging power by (17), and the bus rises briefly by +3.5% before the voltage loop and the supercapacitor absorb the surplus. At 12:40 the PV converter enters curtailment and about 30 kW of PV power is removed in one step, which

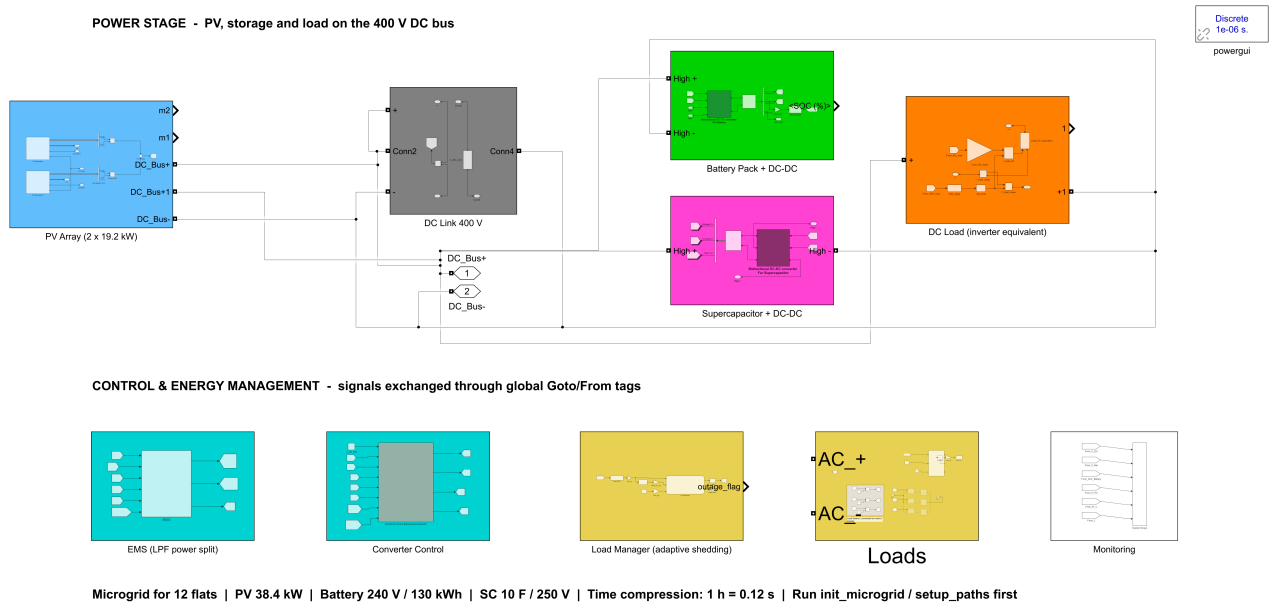


Fig. 12. Top level of the integrated Simulink model (MATLAB R2025a). Upper row: power stage on the 400 V DC bus. Lower row: EMS, converter control, Stateflow load manager, load profile and monitoring.

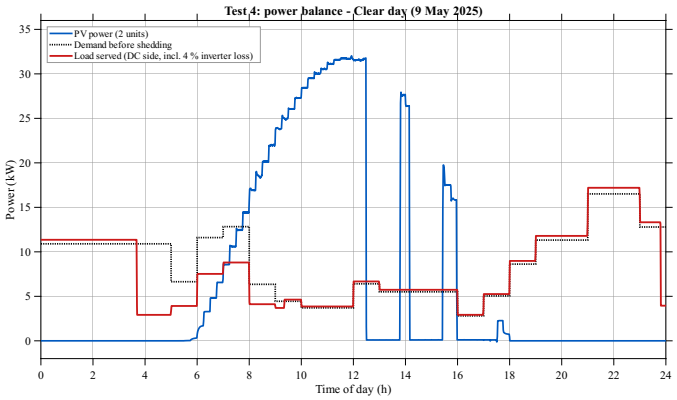


Fig. 13. Test 4, clear day: two-unit PV power, demand before shedding, and load served on the DC side (including the 4% inverter loss).

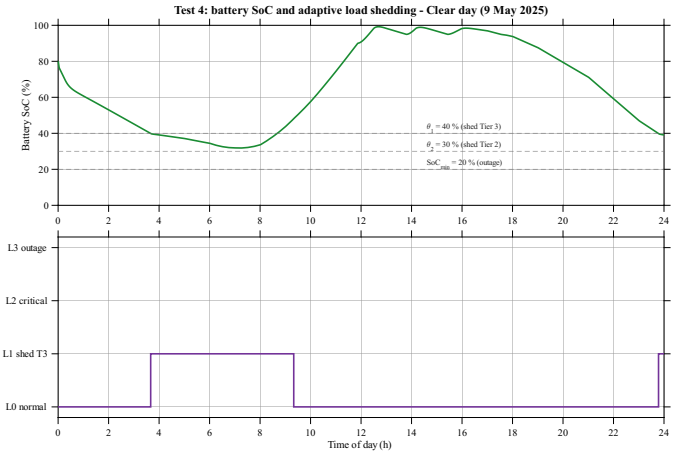


Fig. 14. Test 4, clear day: battery SoC with the shedding thresholds (top) and load-manager state L (bottom).

causes the largest dip of -1.9%; the smaller peaks at 14:00 and 15:30 occur when MPPT resumes. Fig. 16 shows how the HESS shares the work. The battery current is smooth and carries the sustained power, while the supercapacitor voltage stays between about 249.8 and 250.3 V and moves only at the PV switch-overs, where it supplies or absorbs the fast part of the step. This is the intended split of (19). The battery terminal voltage stays between 262 and 277 V,.

Partly cloudy day. Up to 06:00 the run is identical to the clear day, because the PV output is zero and the initial SoC is the same, so Tier 3 is again shed at 03:40. With less irradiance in the morning the battery starts charging later, the SoC falls to about 30% near 07:50, just above θ_2, and Tier 3 is restored only at 10:11, 51 minutes later than on the clear day (Fig. 17). The two PV units peak at 26.4 kW (Fig. 18), the battery reaches 98% only at about 15:00, and PV is curtailed once. The DC bus stays between 395.7 V and 406.9 V (1.7%), and the day ends at an SoC of 38.3%. On both days only Tier 3 is shed; the critical-only state $L = 2$ and the outage state $L = 3$ are never entered, so the critical and essential loads are supplied throughout.

Storage response on the partly cloudy day. Fig. 19 shows the SoC and the load-manager state on the partly cloudy day, and Figs. 20 and 21 show the DC bus and the HESS. Because the PV power rises more slowly, the battery current crosses from discharge to charge at about 07:50, about 30 minutes later than on the clear day, and the charging current peaks at about 75 A near noon instead of 100 A. The SoC reaches 90% at about 13:45. At this instant the EMS derates the charging power, the battery current steps from about -40 A towards zero,

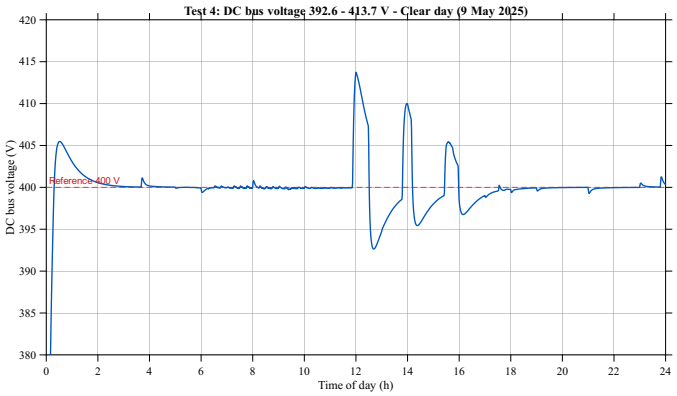


Fig. 15. Test 4, clear day: DC-bus voltage over 24 hours.

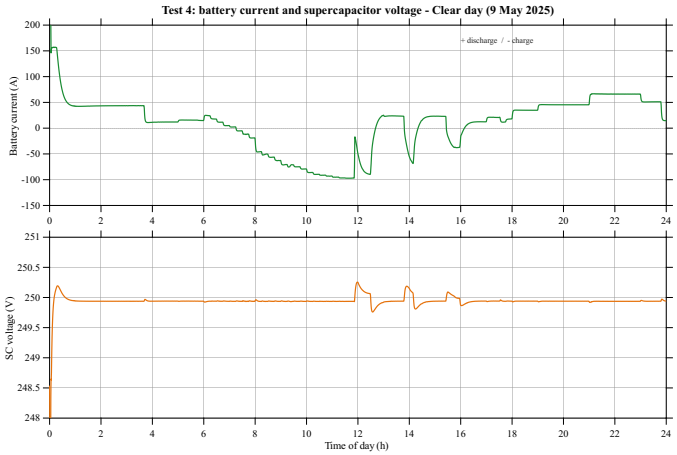


Fig. 16. Test 4, clear day: battery current (positive for discharge) and supercapacitor voltage.

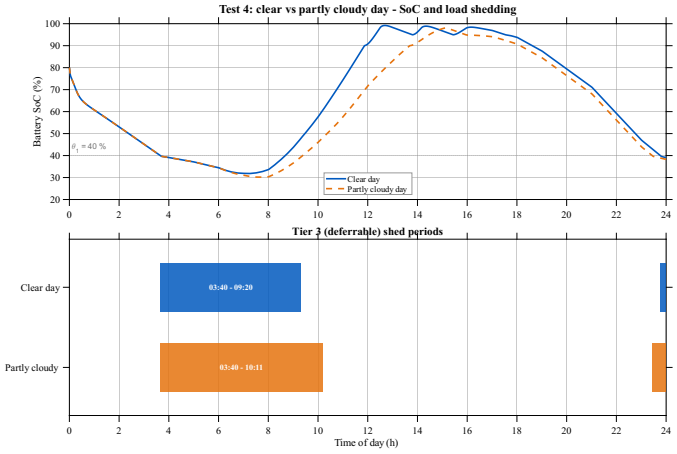


Fig. 17. Tests 3 and 4: battery SoC on the clear and partly cloudy days (top) and the periods in which Tier 3 is shed (bottom).

and the bus rises to 406.9 V while the supercapacitor voltage moves by less than 0.2 V. At about 15:05 the SoC reaches 98%, PV is curtailed once, and the bus dips to 395.7 V; because the irradiance is already falling, MPPT is not resumed with a large step and no second curtailment cycle occurs. Throughout

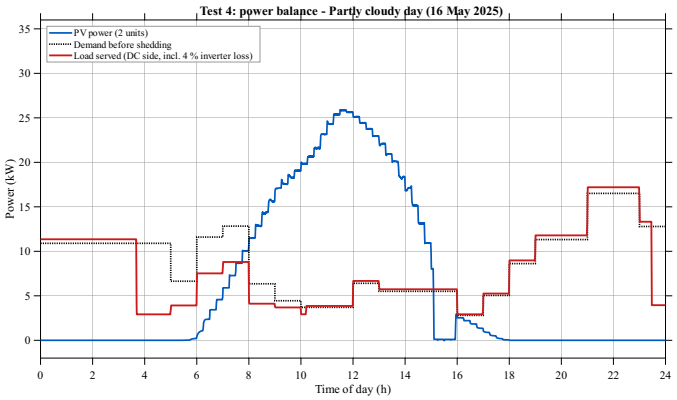


Fig. 18. Test 4, partly cloudy day: two-unit PV power, demand before shedding, and load served on the DC side.

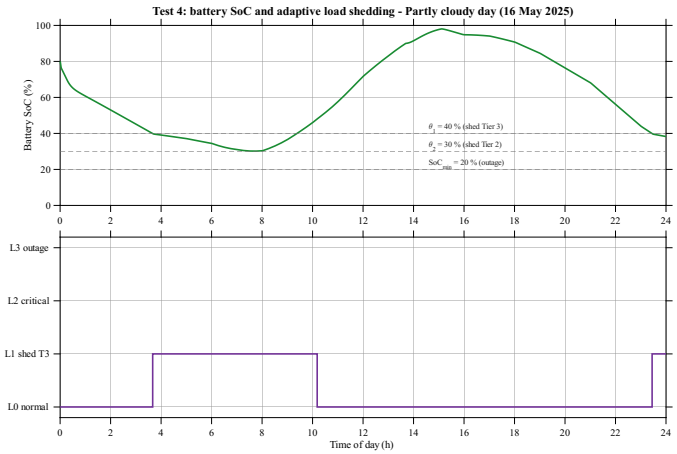


Fig. 19. Test 4, partly cloudy day: battery SoC with the shedding thresholds (top) and load-manager state L (bottom).

the day the supercapacitor voltage stays within about 0.2 V of 250 V and moves only at these two events, while the battery current follows the slow trend of the load and PV, which confirms that the low-pass split of (19) works under variable irradiance as well as on the clear day. The bus deviation on the partly cloudy day (1.7%) is smaller than on the clear day (3.5%) because the PV steps at curtailment are smaller.

Daily energy. Both days end near 39%, below the 80% at which they start, although the clear-day PV energy exceeds the demand. Two effects cause this. Between 12:30 and 16:00 the battery is full and most of the surplus is curtailed, and from 16:00 to 07:00 the block demands about 150 kWh with Tier 3 connected, far more than the 20–90% battery window holds. The battery was sized for the firm night demand, so shedding Tier 3 at night is the designed response rather than a failure. In a sequence of days, however, the next day would start near 39% and Tier 3 would be shed for most of the night. Moving deferrable loads such as water heating, pumping and laundry into the curtailment period would use the surplus that is now wasted.

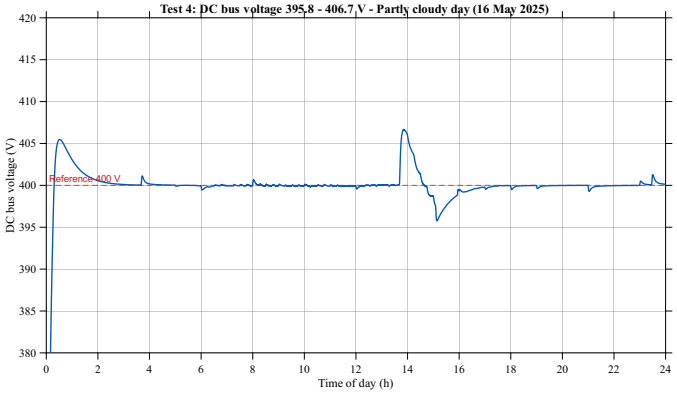


Fig. 20. Test 4, partly cloudy day: DC-bus voltage over 24 hours.

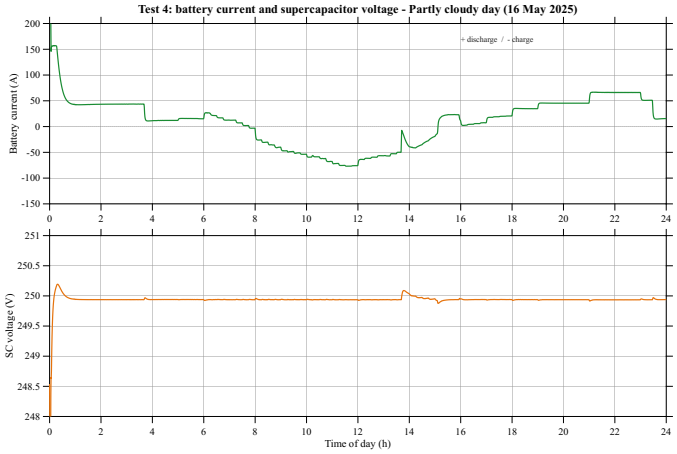


Fig. 21. Test 4, partly cloudy day: battery current (positive for discharge) and supercapacitor voltage.

TABLE X
PERFORMANCE INDICES FOR THE TESTED CASES

Index	Value	Source
Peak irradiance, clear / cloudy	1002 / 790 $\mathrm{W/m^2}$	Weather
Peak cell temperature (clear)	66.6 °C	Test 1
Peak PV power per unit	16.0 / 13.1 kW	Test 1
PV voltage / duty ratio near noon	197 V / 0.51	Test 1
PV energy, two units	238.9 / 155.3 kWh	Test 1
Daily demand, total / firm	211.6 / 117.8 kWh	Load model
Coverage ratio ρ, clear / cloudy	1.13 / 0.73	Test 1
Inverter output	225.8 V, 19.1 A RMS	Test 2
Peak output voltage	325.7 V	Test 2
Voltage THD to 50th / to 10 kHz	0.13% / 1.57%	Test 2
DC bus, clear day	392.6–413.9 V	Test 4
DC bus, partly cloudy day	395.7–406.9 V	Test 4
Tier 3 shed, clear / cloudy	03:40–09:20 / 03:40–10:11	Tests 3, 4
Minimum SoC, clear / cloudy	about 32% / 30%	Test 4
SoC at 24:00, clear / cloudy	39.1% / 38.3%	Test 4
Tier 1 and 2 interruptions	None	Tests 3, 4

E. Limitations

The time-compressed model has the following limitations:

- Start-up transients that last milliseconds are booked as real hours and cost about 7% SoC at the start of each run.
- The battery block keeps the real capacity, so its terminal voltage follows the block's own, almost constant SoC rather than the compressed SoC of (11); the voltage does not show the end-of-discharge drop.
- Each day starts from an assumed SoC of 80%, and both days end near 39%; a multi-day run is needed to find the periodic operating point.
- In curtailment the PV output falls almost to zero instead of supplying the load, so part of the surplus is wasted.
- The integrated model represents the inverter by an equivalent DC load with resistive (unity power factor) loads. A physical CHB inverter also needs an isolated DC source for each cell, which requires isolated DC–DC stages from the 400 V bus.
- Hourly weather data, interpolated to one minute, smooths out fast cloud transients, and the horizontal irradiance is used without transposition to a tilted array.

VII. CONCLUSION

This paper presented a MATLAB/Simulink design of a solar-powered microgrid for 12 flats, combining a 38.4 kW PV array with P&O MPPT, a 400 V DC bus, battery–supercapacitor storage with SoC-aware low-pass-filter sharing, a priority load manager and a five-level CHB inverter. With NASA POWER weather for Coimbatore, one PV unit peaks at 16.0 kW, in agreement with the temperature-corrected estimate, and the two units supply 238.9 kWh on a clear day and 155.3 kWh on a partly cloudy day, 113% and 73% of the 211.6 kWh demand. The five-level inverter delivers 226 V RMS with 0.13% voltage THD up to the 50th harmonic.

The integrated model runs over complete days in closed loop. The DC bus stays within 392.6–413.9 V, the battery carries the sustained power while the supercapacitor takes the fast steps, PV is curtailed cleanly when the battery is full, and the load manager sheds only the deferrable tier, from 03:40 to 09:20 on the clear day and to 10:11 on the partly cloudy day, without interrupting critical or essential loads.

The results also confirm the sizing: the 38.4 kW array covers the firm demand even on the partly cloudy day, the 130 kWh battery carries the firm night demand, and the 10 F supercapacitor, storing only about 0.09 kWh, absorbs every fast power step with a voltage change below 0.5 V.

Future work will run several consecutive days to find the periodic SoC, schedule deferrable loads into the midday surplus instead of curtailing it, let the PV converter share the load during curtailment, model reactive (RLC) loads and the isolated DC–DC stages of the CHB inverter, and validate the controller in hardware; grid-connected operation would also need to meet IEEE Std 1547 [28].

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